Anteater Chess Game (Version 1.2)

By Thoughtless Airheads

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Glossary of Anteater Chess Terms

Game

- **Anteater Chess**–a variation of the game chess in which pawns are replaced by pieces named “ants” and players have additional pieces named “anteaters”; the board configuration and starting layouts are changed in response to these changes (see “Ranks”, “Files”, and “Anteater”)
- **Board**–the interface through which the game is played; rectangular setup of squares that can be occupied by pieces; size determined by the number of ranks and files (see “Ranks” and “Files”)
- **Check**–one player is capable of capturing their opponent’s king on the player’s next move, but the opponent is capable of making a move such that their king cannot be captured during the player’s next move
- **Checkmate**–one player is guaranteed to have their king captured during the opponent’s next move, regardless of their own move; this ends the game, with the aforementioned opponent then being declared the winner of the game
- **Files**–the columns of the board, labeled from the left side of the board to the right; while there are eight (labeled A through H) files in normal chess, there are 10 (labeled A through J) files in anteater chess
- **Piece**–a general term used to refer to a(n) ant, anteater, bishop, king, knight, queen, or rook
- **Ranks**–the rows of the board, numbered from the bottom of the board to the top; there are eight (numbered 1 through 8) ranks in both normal chess and anteater chess
- **Turn**–a player moves one piece to a desired, legal position; turns alternate between players, starting with the player using the white chess set

Pieces

- **Ant**–the equivalent of a pawn in a normal chess game; usually moves one square forward at a time, but can also move two squares forward at a time if it is the very first move performed by a player in the game, and can move diagonally to capture pieces (see “Capture”)
    - Each player has 10 of these at the beginning of the game
- **Anteater**–can move one square in any direction like a king; cannot capture any pieces except ants (i.e. not a threat to the opposing king), but can capture any and all ants adjacent to the a captured ant, provided that they are connected vertically or horizontally (see “Ant eating”); does not have an equivalent in a normal chess game
    - Each player has two of these at the beginning of the game
- **Bishop**–behaves the same as a bishop in a regular chess game; can move diagonally along an unobstructed path
    - Each player has two of these at the beginning of the game

- **King**–behaves the same as a king in a regular chess game; can move one square in any direction, but can never move into check (see “Check”)
    - Each player has one of these at the beginning of the game
- **Knight**–behaves the same as a knight in a regular chess game; can move two squares in a straight line and then one square in a perpendicular direction; can “jump” over pieces (i.e. does not require an unobstructed path)
    - Does not capture pieces it “jumps” over
    - Each player has two of these at the beginning of the game
- **Queen**–behaves the same as a queen in a regular chess game; can move in any direction (horizontally, vertically, and diagonally) along an unobstructed path
    - Each player has one of these at the beginning of the game
- **Rook**–behaves the same as a rook in a regular chess game; can move in a straight line (horizontally and vertically) along an unobstructed path
    - Each player has two of these at the beginning of the game

Moves

- **Ant eating**—special capture move that can be performed by the anteater piece (see “Anteater”); can capture any and all ants adjacent to a captured ant, provided they are connected vertically or horizontally; anteater piece ends in the position of the last eaten ant, with the order of “eating” possibly up to the player
    - Note: Player can choose to not “eat” all reachable ants during their move
- **Capture**–the player moves a piece to the position of an opponent’s piece (except for en passant; see “En Passant”) and takes the opponent’s piece off the board; the removed piece is then considered out of play
- **Castling**—There are two conditions to use castling. First, King and Rook should be in their original position(must have not moved before). Second, no piece can be between King and Rook. Third, the King must not currently be in check. Fourth, the King cannot pass through or land on a square that is under attack. Once all conditions are satisfied, the player can move King two squares towards Rook, and then place Rook on the square next to the king on the opposite side.
- **En passant**—A pawn capture that can occur when an opponent’s pawn moves two squares forward from its starting position and lands beside your pawn. If your pawn is on the fifth rank (for White) or fourth rank (for Black), you may capture the opponent’s pawn as if it had only moved one square forward. This capture must be made immediately on the next move, otherwise the opportunity is lost.
- **(Moving) backward**–a piece is moved towards its team’s starting position on the board
- **(Moving) forward**–a piece is moved away from its team’s starting position on the board
- **Pawn promotion**–any pawn piece that reaches the rank furthest from their starting position immediately exchanges their pawn for a Queen, Rook, Bishop, Knight, or Anteater; the new piece does not have to be one that was previously captured

1. Overview and Tutorial

1.1 Anteater Chess Rules/Gameplay

The user should first open the program by calling the executable file as specified in Section 2. Upon opening the program, the user will see a starting menu.

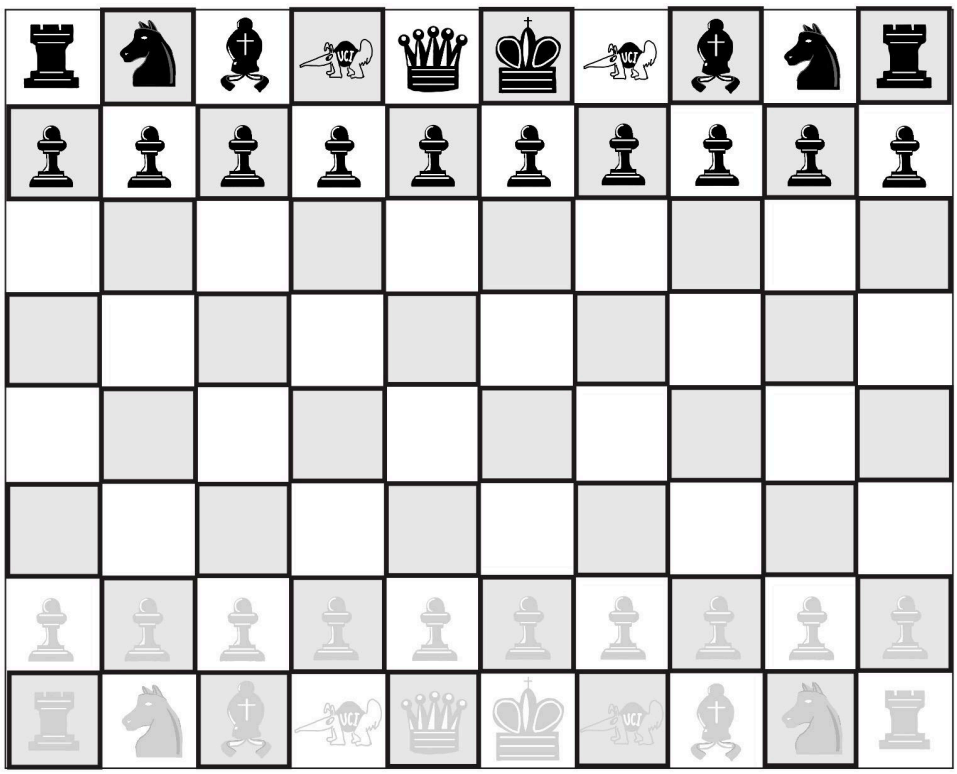
If the user chooses to start a game, the user will choose between the white set of pieces and the black set of pieces; the opponent uses the other.

The player using the white set of pieces starts first. Players alternate taking turns, using only legal moves, until one player's king is captured or forced into checkmate. The player who captured the king or imposed a checkmate on the other player wins. If there are no legal moves but both kings are not in check(stalemate), agreement between players.

After completion of the game, the player will return to the starting menu.

1.2 Usage Scenarios

Sketch of a typical screenshot (GUI, final version):



Making a move

```
Select what piece you want to move: E2
Select where you want to go: E4
  +----+----+----+----+----+----+----+----+----+----+
8 | bR | bN | bB | bA | bQ | bK | bA | bB | bN | bR |
  +----+----+----+----+----+----+----+----+----+----+
7 | bP | bP | bP | bP | bP | bP | bP | bP | bP | bP |
  +----+----+----+----+----+----+----+----+----+----+
6 |    |    |    |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
5 |    |    |    |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
4 |    |    |    |    | wP |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
3 |    |    |    |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
2 | wP | wP | wP | wP |    | wP | wP | wP | wP | wP |
  +----+----+----+----+----+----+----+----+----+----+
1 | wR | wN | wB | wA | wQ | wK | wA | wB | wN | wR |
  +----+----+----+----+----+----+----+----+----+----+
    A    B    C    D    E    F    G    H    I    J
```

Illegal Move by User

```
  +----+----+----+----+----+----+----+----+----+----+
8 | bR | bN | bB | bA | bQ | bK | bA | bB | bN | bR |
  +----+----+----+----+----+----+----+----+----+----+
7 | bP | bP | bP | bP | bP | bP | bP | bP | bP | bP |
  +----+----+----+----+----+----+----+----+----+----+
6 |    |    |    |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
5 |    |    |    |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
4 |    |    |    |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
3 |    |    |    |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
2 | wP | wP | wP | wP | wP | wP | wP | wP | wP | wP |
  +----+----+----+----+----+----+----+----+----+----+
1 | wR | wN | wB | wA | wQ | wK | wA | wB | wN | wR |
  +----+----+----+----+----+----+----+----+----+----+
    A    B    C    D    E    F    G    H    I    J
```

Select what piece you want to move: **E2**
Select where you want to go: **D4**
Not a legal move, please choose again.
Select what piece you want to move:

User won the game

```
Select what piece you want to move: E5
Select where you want to go: E6
  +----+----+----+----+----+----+----+----+----+----+
8 |    |    |    |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
7 |    |    |    | bP |    |    |    |    | wP | bP |
  +----+----+----+----+----+----+----+----+----+----+
6 |    |    |    |    | bQ |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
5 |    | bP | bB |    |    |    |    | wA |    |    |
  +----+----+----+----+----+----+----+----+----+----+
4 |    |    |    |    |    |    |    | wP |    |    |
  +----+----+----+----+----+----+----+----+----+----+
3 |    |    |    | wP |    |    |    |    |    | wP |
  +----+----+----+----+----+----+----+----+----+----+
2 |    |    |    | wK |    | wR | bR |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
1 | wR |    |    |    |    |    |    |    | wB |    |
  +----+----+----+----+----+----+----+----+----+----+
    A    B    C    D    E    F    G    H    I    J

White wins!
```

Computer won the game

```
Select what piece you want to move: F1
Select where you want to go: F2
  +----+----+----+----+----+----+----+----+----+----+
8 | bR | bN | bB | bA | bQ | bK | bA |    | bN | bR |
  +----+----+----+----+----+----+----+----+----+----+
7 | bP | bP | bP | bP | bP | bP | bP | bP |    | bP |
  +----+----+----+----+----+----+----+----+----+----+
6 |    |    |    |    |    |    |    |    | bP | bB |
  +----+----+----+----+----+----+----+----+----+----+
5 |    |    |    |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
4 |    |    | wP |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
3 |    |    |    |    |    | wP |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
2 | wP | wP |    | wP | wP | wK | wP | wP | wP | wP |
  +----+----+----+----+----+----+----+----+----+----+
1 | wR | wN | wB | wA | wQ |    | wA | wB | wN | wR |
  +----+----+----+----+----+----+----+----+----+----+
    A    B    C    D    E    F    G    H    I    J

Black moving...
Black moves J6 to F2
```

```
  +----+----+----+----+----+----+----+----+----+----+
8 | bR | bN | bB | bA | bQ | bK | bA |    | bN | bR |
  +----+----+----+----+----+----+----+----+----+----+
7 | bP | bP | bP | bP | bP | bP | bP | bP |    | bP |
  +----+----+----+----+----+----+----+----+----+----+
6 |    |    |    |    |    |    |    |    | bP |    |
  +----+----+----+----+----+----+----+----+----+----+
5 |    |    |    |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
4 |    |    | wP |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
3 |    |    |    |    |    | wP |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
2 | wP | wP |    | wP | wP | bB | wP | wP | wP | wP |
  +----+----+----+----+----+----+----+----+----+----+
1 | wR | wN | wB | wA | wQ |    | wA | wB | wN | wR |
  +----+----+----+----+----+----+----+----+----+----+
    A    B    C    D    E    F    G    H    I    J
```

Black wins!

Draw

```
Select what piece you want to move: E4
Select where you want to go: E5
  +----+----+----+----+----+----+----+----+----+----+
8 |    |    |    |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
7 |    |    |    |    | bK |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
6 |    |    |    |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
5 |    |    |    | wP | wK |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
4 |    |    |    |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
3 |    |    |    |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
2 |    |    |    |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
1 |    |    |    |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
    A    B    C    D    E    F    G    H    I    J

It’s a draw!
```

Time out

```
Select what piece you want to move: E2
Select where you want to go: E4
  +----+----+----+----+----+----+----+----+----+----+
8 | bR | bN | bB | bA | bQ | bK | bA | bB | bN | bR |
  +----+----+----+----+----+----+----+----+----+----+
7 | bP | bP | bP | bP | bP | bP | bP | bP | bP | bP |
  +----+----+----+----+----+----+----+----+----+----+
6 |    |    |    |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
5 |    |    |    |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
4 |    |    |    |    | wP |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
3 |    |    |    |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
2 | wP | wP | wP | wP |    | wP | wP | wP | wP | wP |
  +----+----+----+----+----+----+----+----+----+----+
1 | wR | wN | wB | wA | wQ | wK | wA | wB | wN | wR |
  +----+----+----+----+----+----+----+----+----+----+
    A    B    C    D    E    F    G    H    I    J

Black moving...
Time’s up!
White wins!
```

User Log (Algebraic Notation)

White	Black
1. e2-e4	1. e7-e5
2. Kt g1-f3	2. Kt b8-c6
3. B f1-b5	3. a7-a6
4. B b5-a4	4. Kt g8-f6
5. o-o	5. b7-b5
6. B a4-b3	6. B f8-e7
7. d2-d4	7. d7-d6
8. c2-c3	8. B c8-g4
9. B c1-e3	9. o-o

1.3 Goals

To create a fully functional chess game capable of detecting legal moves, raising the necessary error messages, and updating the chess board correctly based upon the collective moves made by both the computer and user.

1.4 Features

The program will be able to support the following gamemodes with

- Player vs Computer
- Player vs Player
- Computer vs Computer

Additional features:

- Undo current move (for the player)
- Help menu
    - Scroll down text that explains how to play and use the program.
- (potentially) providing clock timers for both players
- Separate log file in algebraic notation
- TBD

Reprompt them if the user types out the move in the wrong format

2. Installation

2.1 System requirements

The user must satisfy the following conditions to correctly install the software:

- The user must have enough storage space on their computer to download the program
- The user must be able to access and use the command line to start the program
- The user must have RHEL8 64-bit Linux

2.2 Setup and configuration

Download the package from the website AmesomeSoft.com. Using the command line, go into the installed folder.

For the binary package, run the executable program on the command line as follows:

```
% gtar xvzf Chess_V1.0.tar.gz (extract contents into current directory)
% evince chess/doc/chess.pdf (for reading documentation, if desired)
% chess/bin/chess (run executable file)
```

For the source package, build the software using the following command line instructions (more details are in parentheses):

```
% gtar xvzf Chess_V1.0_src.tar.gz (extract contents into current
directory)
% evince chess/doc/chess\_software.pdf (for reading documentation, if
desired)
% cd chess (go into chess directory)
% make (make executable file)
% make test (for testing code, if desired)
% make clean (to remove executables once finished, if desired)
```

(More detailed/accurate instructions TBD.)

2.3 Uninstalling

Through the command line, the user can uninstall the program by calling *make clean* (see above), then going one directory above the installed directory and calling *rmdir -rf ChessDirectoryName*, with ChessDirectoryName being the name of the directory that was

downloaded by the user (the user may have decided to rename the directory, so be sure to check its current name).

(More detailed/accurate instructions TBD.)

3. Functions and Features

The full list of diagrams can be found under “Section 1.2 Usage Scenarios”.

3.1 User interface

- Starting menu
  - The user can choose to start a game or exit the program, and the program will update accordingly based on this input. The user can choose whether the game is computer vs. player, player vs. player, and computer vs. computer (see image below).
- Game interface
  - If the user chooses to start a game, they will see a text display of the board (alpha version; see image below) or a GUI (final version). The user will then receive instructions for how to make moves.
    - Note: For both versions, user input will be text-based.

```
Welcome to Anteater Chess!
Choose your Gamemode:
   1. Computer vs Player
   2. Player vs Player
   3. Computer vs Computer
Make your pick: 1
Choose your color (white/black): white
Good Luck!

  +----+----+----+----+----+----+----+----+----+----+
8 | bR | bN | bB | bA | bQ | bK | bA | bB | bN | bR |
  +----+----+----+----+----+----+----+----+----+----+
7 | bP | bP | bP | bP | bP | bP | bP | bP | bP | bP |
  +----+----+----+----+----+----+----+----+----+----+
6 |    |    |    |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
5 |    |    |    |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
4 |    |    |    |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
3 |    |    |    |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
2 | wP | wP | wP | wP | wP | wP | wP | wP | wP | wP |
  +----+----+----+----+----+----+----+----+----+----+
1 | wR | wN | wB | wA | wQ | wK | wA | wB | wN | wR |
  +----+----+----+----+----+----+----+----+----+----+
    A    B    C    D    E    F    G    H    I    J
```

3.2 Playing Modes

- This software supports the following modes, to be selected by the user in the starting menu (see above):
    - Player vs. computer
    - Player vs. player
    - Computer vs. computer
- Currently, this software only has one difficulty mode for the computer player

```
Welcome to Anteater Chess!
Choose your Gamemode:
   1. Computer vs Player
   2. Player vs Player
   3. Computer vs Computer
Make your pick: 1
Choose your color (white/black): white
Good Luck!

  +----+----+----+----+----+----+----+----+----+----+
8 | bR | bN | bB | bA | bQ | bK | bA | bB | bN | bR |
  +----+----+----+----+----+----+----+----+----+----+
7 | bP | bP | bP | bP | bP | bP | bP | bP | bP | bP |
  +----+----+----+----+----+----+----+----+----+----+
6 |    |    |    |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
5 |    |    |    |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
4 |    |    |    |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
3 |    |    |    |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
2 | wP | wP | wP | wP | wP | wP | wP | wP | wP | wP |
  +----+----+----+----+----+----+----+----+----+----+
1 | wR | wN | wB | wA | wQ | wK | wA | wB | wN | wR |
  +----+----+----+----+----+----+----+----+----+----+
    A    B    C    D    E    F    G    H    I    J
```

3.3 Move Logs

- The program will produce a text file (.txt) that contains a human-readable log of moves (official algebraic notation) made during the game by both Black and White players (ex. 01. e2-e4). The user will be able to open this text file via the command line or whatever other options they have for opening .txt files. See image below for an example.

(algebraic notation)

White	Black
1. e2-e4	1. e7-e5
2. Kt g1-f3	2. Kt b8-c6
3. B f1-b5	3. a7-a6
4. B b5-a4	4. Kt g8-f6
5. o-o	5. b7-b5
6. B a4-b3	6. B f8-e7
7. d2-d4	7. d7-d6
8. c2-c3	8. B c8-g4
9. B c1-e3	9. o-o

Troubleshooting: Error Messages

Differing Scenarios:

- If the move made by the user wasn't legal: "User move was illegal: please make an allowed move"
- If the game doesn't detect that the game was won: "Game fails to recognize game was won…Exiting Program"
- if the game fails to install correctly: "Game failed to install [include missing files]"
- The computer fails to provide a move: "Computer made no move…Exiting Program"
- The board fails to update correctly: "Reloading board"
- The computer provides an illegal move: "Computer provided an illegal move…Exiting Program"
- Program times out: "Program timed out…Exiting Program"
- The players don't switch (black to white or white to black): "Players failed to switch turns…Exiting Program"
- A taken piece returns to the board: "A previous piece returned to the board not by Pawn Promotion…Exiting Program"
- Game freeze or crash: "Issues with making move within allotted time…Please exit program"
- etc.

(Error messages are subject to change)
TBD based upon coding

Contact Information

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